Adrian Kieback

AI Researcher: Speech Foundation Models and AI for Mental Health

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RESEARCH PROFILE

AI researcher working where speech foundation models meet mental health care. My research builds interpretable, compute-efficient speech AI that turns a short clinical interview into objective, auditable markers of psychiatric symptom severity, in a field where assessment still depends on subjective rating scales. I co-developed S-JEPA, a self-supervised speech model created with collaborators at NYU (Yann LeCun, Ravid Shwartz-Ziv), Carnegie Mellon, Columbia and Stanford, and applied it in a clinical study of pediatric bipolar disorder with Baylor College of Medicine and Texas Children's Hospital, published in the *Journal of the American Academy of Child and Adolescent Psychiatry*. Related work spans speech-based depression detection, wearable sensing for lung cancer screening, and reliable agentic AI systems.

Research areas: AI for mental health and speech biomarkers • self-supervised speech representation learning • interpretable and explainable clinical AI • digital health and wearable sensing • agentic LLM systems

RESEARCH APPOINTMENTS

AI Engineer and Researcher 08/2026 - Present
Discern Science International, Inc. *Tucson, AZ*

- Building the company's mental health research line: speech-based assessment of bipolar disorder and related conditions, continuing the collaboration with San Diego State University, Baylor College of Medicine and Texas Children's Hospital.
- Develop speech and language AI for the company's structured interview analytics platform, a decision-support system in which authorized human reviewers make every decision.
- Built, with my team, an in-house avatar interviewer pipeline (text-to-speech and audio-driven lip-synced video generation) that replaced an external API, so structured interviews run automatically on the company's own infrastructure.

Machine Learning Researcher 09/2023 - 08/2026
James Silberrad Brown Center for AI, San Diego State University *San Diego, CA*

- Conducted research on speech-based mental health assessment, self-supervised speech representation learning, clinical digital health and agentic AI, resulting in seven papers, abstracts and preprints (see Publications and Research Contributions).
- Led research teams of two to four engineers and clinical collaborators and served as technical lead on studies with Sharp HealthCare and on an NSF I-Corps team.

EDUCATION

Master of Science, Big Data Analytics 08/2022 - 05/2024

San Diego State University *San Diego, CA*
GPA 3.92. Thesis: "Enhancing Mental Health Diagnosis with Representation Learning: A Transformer-Based Approach."

Bachelor of Science, Business Informatics 10/2018 - 07/2022
Albstadt-Sigmaringen University *Germany*
GPA 3.46. Thesis: "Conceptual design of a central tool for the globally standardized presentation and analysis of process and machine data" (written at Carl Zeiss AG).

PUBLICATIONS

Citation metrics (11/2026): 26 citations (Google Scholar); h-index 3 and 4 highly influential citations (Semantic Scholar). * denotes equal contribution.

Peer-Reviewed Papers

[1] **Kieback, A.**, Amadasun, U. P., Chadha, A., & Elkins, A. (2026). TASR: Training-Free Adaptive Stopping for Iterative Retrieval. *Agent4IR Workshop at KDD 2026*; to appear as a chapter in *Communications in Computer and Information Science* (Springer). arXiv:2606.13814. Lead and corresponding author.

[2] Elkins, A. C., Singh, S., **Kieback, A.**, Blankenship, S., Amadasun, U. P., & Chadha, A. (2025). DAWZY: A New Addition to AI Powered "Human in the Loop" Music Co-creation. *NeurIPS 2025 Workshop on AI for Music* (poster). arXiv:2512.03289.

[3] Elkins, A. C., Singh, S., Blankenship, S., **Kieback, A.**, Amadasun, U. P., & Chadha, A. (2025). DAWZY: Human in the Loop Natural Language Control of REAPER. *NeurIPS 2025 Workshop on AI for Music* (demo).

Journal-Published Conference Abstracts

[4] Elkins, A., **Kieback, A.**, Chadha, A., Kurian, S., Burgoon, J. K., Dunbar, N. E., Simonetti, A., Walls, B., & Saxena, K. (2026). Listening for Mania: Interpretable Speech Clusters Track Symptom Severity in Pediatric Bipolar Disorder. *Journal of the American Academy of Child and Adolescent Psychiatry*, 65(10S), S270 (abstract 3.47). doi:10.1016/j.jaac.2026.08.232.

Preprints

[5] Ioannides, G., **Kieback, A.***, Goldfeder, J.*, Pang, L., Chadha, A., Elkins, A., LeCun, Y., & Shwartz-Ziv, R. (2026). S-JEPA: Soft Clustering Anchors for Self-Supervised Speech Representation Learning. arXiv:2606.19398.

[6] Ioannides, G., **Kieback, A.**, Goldfeder, J., Pang, L., Chadha, A., Elkins, A., LeCun, Y., & Shwartz-Ziv, R. (2026). Soft Clustering Anchors for Self-Supervised Speech Representation Learning in Joint Embedding Prediction Architectures. arXiv:2602.09040.

[7] Ioannides, G.*, **Kieback, A.***, Chadha, A., & Elkins, A. (2024). Density Adaptive Attention-based Speech Network: Enhancing Feature Understanding for Mental Health Disorders. arXiv:2409.00391.

CONFERENCE PRESENTATIONS

- **AACAP 2026**, Annual Meeting of the American Academy of Child and Adolescent Psychiatry, Atlanta, GA. Poster presenter: "Listening for Mania: Interpretable Speech Clusters Track Symptom Severity in Pediatric Bipolar Disorder."

- **WM@Booth 2026**, World Models Workshop, University of Chicago Booth School of Business, Chicago, IL. Poster presenter: S-JEPA.
- **NeurIPS 2025**, Workshop on AI for Music. Presented DAWZY (poster and live demo).

PEER REVIEW AND PROFESSIONAL SERVICE

- Reviewer, Workshop on Structured Data for Health (SD4H) at the International Conference on Machine Learning (ICML) 2026.

RESEARCH CONTRIBUTIONS

AI for Mental Health: Objective Speech Markers of Symptom Severity

Listening for Mania: speech markers of mania in pediatric bipolar disorder
With Baylor College of Medicine, Texas Children's Hospital, University of Arizona, UC Santa Barbara and Fondazione Policlinico Gemelli (Rome). Funded by the Milken Institute.

- Pediatric bipolar disorder is frequently misdiagnosed and has no objective measure of mood state. This study tests whether a short structured voice interview can provide one.
- Encoded interview speech from 30 youth (ages 8 to 17; 15 with bipolar disorder, 15 healthy controls) with S-JEPA at 20 ms resolution and grouped it into 16 interpretable "speech types" with a Gaussian mixture model, using no diagnostic labels.
- Identified a speech cluster that tracks clinician-rated mania severity (YMRS; $r = 0.42$, 95% CI 0.14 to 0.69) and was nearly absent in 14 of 15 healthy controls, and a second cluster associated with milder symptoms ($r = -0.36$, 95% CI -0.54 to -0.12).
- Unlike score-only AI tools, the method shows clinicians which parts of a child's speech were flagged. Pilot results; replication in a larger sample is the next step.

Speech-based depression detection with density adaptive attention
Master's thesis research and follow-up paper; equal first author. With Carnegie Mellon and Stanford collaborators.

- Led an end-to-end depression detection project on the DAIC-WOZ clinical interview corpus, from the audio pipeline through statistically rigorous evaluation and interpretability analysis.
- Exceeded the previous best speech-only result (SpeechFormer, macro F1 0.694 with 33.2M parameters) using a 280K-parameter model (macro F1 0.702), about 1% of the parameters, small enough for clinical and on-device deployment.
- Reached macro F1 0.72 with a 1.1M-parameter Transformer, the best result among compared models trained on depression labels only.
- Made predictions explainable: the attention mechanism exposes the frequency bands that drive each decision, including the fundamental pitch range known to change in depression.

Speech Foundation Models

S-JEPA: soft clustering anchors for self-supervised speech representation learning
Equal contribution. With collaborators at NYU, Carnegie Mellon, Columbia, Northeastern and Stanford.

- Co-developed a self-supervised speech encoder that replaces the hard k-means targets of HuBERT-style models with soft Gaussian mixture posteriors updated online, which preserves acoustic ambiguity at phoneme boundaries and removes the offline re-clustering step.

- Pretrained the 51.8M-parameter model on approximately 83,000 hours of English speech using eight NVIDIA B200 GPUs.
- Achieved the lowest word error rate among self-supervised models under 90M parameters on LibriSpeech (12.10%) and matched HuBERT Base on emotion recognition (64.83% vs. 64.92%) with 55% of its parameters (SUPERB benchmark, frozen encoder).
- The model's representations are the basis of the pediatric bipolar disorder study above, linking foundation model research directly to clinical use.

Digital Health and Clinical Data Science

Wearable smoking detection for lung cancer screening
Technical lead. Pilot study of the SDSU/UC San Diego U54 CREATE Partnership (NIH/NCI U54CA285117) with Sharp HealthCare.

- Lead the engineering of an iOS research platform (ResearchKit, HealthKit, SensorKit) that combines Apple Watch passive sensing, surveys and electronic health records to detect smoking behavior objectively, since screening eligibility currently rests on unreliable self-reported smoking history.
- Led a team of four students using Scrum to build the data pipeline from the ground up, developing them as researchers along the way.
- Apple's SensorKit research entitlement has been granted and the software is complete; the pilot with patient-worn Apple Watches is about to begin. The 12-month study is designed to enroll about 150 adults undergoing lung cancer screening.

Environmental and geographic risk factors for lung cancer
With Sharp HealthCare.

- Build regression, generalized additive and mixed-effects models and geospatial workflows to identify environmental and geographic factors beyond smoking that are associated with lung cancer incidence.

Reliable Agentic AI Systems

TASR: training-free adaptive stopping for iterative retrieval
Lead and corresponding author.

- Designed a training-free rule that tells a retrieval-augmented LLM agent when to stop searching, replacing trained controllers that must be rebuilt for every model and task.
- Improved macro F1 by 3.42 points over fixed three-round retrieval at matched cost and retained 94.8% of the accuracy of five-round retrieval with 62.6% of the LLM calls; one fixed threshold held across 32 model, retriever and corpus configurations.

DAWZY: human-in-the-loop natural language control of audio software
NeurIPS 2025 Workshop on AI for Music.

- Co-developed an open-source assistant that maps text, speech and humming to grounded, reversible edits in a digital audio workstation through Model Context Protocol tools and LLM-generated scripts.
- Reached 44% task success against 0% for a comparable baseline system on matched tasks; rated 4.48 of 5 for enjoyment in a 21-participant user study.

Real-Time Speech Systems and Robotics

Speech-to-speech pipeline for a humanoid robot (Asher)
Technical lead, NSF I-Corps team.

- Engineered a real-time speech-to-speech pipeline for human-robot interaction on an NVIDIA Orin Nano (8 GB), cutting response latency from 2 s to 200 ms, with custom ARM64 builds and automated edge deployment.
- Conducted customer discovery interviews and represented technical feasibility to stakeholders for the NSF I-Corps team.

MENTORING AND TEACHING

Mentor, Aztec Mentor Program 09/2026 - Present
San Diego State University *San Diego, CA*

- Invited by the university to serve as a mentor; guide three students, PhD and undergraduate, on their academic and career path.

Graduate Student Mentor 08/2025 - 08/2026
James Silberrad Brown Center for AI, San Diego State University *San Diego, CA*

- Mentored two graduate capstone teams and one undergraduate researcher working on speech-based mental health modeling, multimodal emotion-aware systems and robustness evaluation.
- Guided model design, experimental methodology and reproducibility, and reviewed proposals, code and results for statistical rigor.

Graduate Assistant 08/2023 - 05/2024
San Diego State University *San Diego, CA*

- Proctored and graded assignments and exams for three undergraduate courses; automated a web-scraping workflow that collected data for faculty research.

ADDITIONAL PROFESSIONAL EXPERIENCE

Data Science Intern 02/2024 - 05/2024
San Diego Association of Governments (SANDAG) *San Diego, CA*

- Automated billing-history checks across more than 90,000 toll customer accounts in support of a billing fraud review, reducing review time from months to minutes.
- Cleaned, validated and visualized geospatial and demographic data for public reporting on the UN Sustainable Development Goals in San Diego County.

Data Analyst Research Assistant 03/2023 - 12/2023
SDSU Research Foundation *San Diego, CA*

- Built reproducible Python and R workflows for data collection, preprocessing and statistical analysis across faculty-led studies.

Business Scientist and Bachelor Thesis Author 02/2022 - 08/2022
Carl Zeiss AG *Aalen, Germany*

- Developed business intelligence dashboards with live data from more than 1,200 manufacturing machines and contributed to a software selection that saved over $30,000 per year.

Project Management Intern 02/2021 - 08/2021
Continental AG *Villingen, Germany*

- Led two change requests for the launch of a fleet management software product and analyzed sales and product-usage data.

LEADERSHIP AND SERVICE

- **Elected Student Council Member and Sports Advisor**, Albstadt-Sigmaringen University, 09/2021 - 09/2022. Represented more than 400 computer science students in departmental decisions and managed the student sports program and its budget.
- **Mathematics Tutor**, Schülerhilfe Albstadt-Ebingen, Germany, 10/2019 - 09/2020.
- **Infrastructure Volunteer**, Kruger National Park, South Africa, 12/2017 - 03/2018. Supported park infrastructure and a camera-trap wildlife monitoring system used for ecological research.

TECHNICAL SKILLS

- **Machine learning:** self-supervised and representation learning, Transformers, probabilistic and mixture models, time-series analysis, uncertainty quantification and calibration, interpretability
- **Speech and audio:** speech foundation model pretraining, log-Mel and spectrogram features, Whisper and Faster-Whisper, text-to-speech, edge deployment on ARM64
- **LLM systems:** retrieval-augmented generation, agentic tool use, Model Context Protocol
- **Programming and infrastructure:** Python, PyTorch, R, SQL, scikit-learn, Hugging Face, Docker, Linux, multi-GPU training, CI/CD
- **Clinical and health data:** iOS ResearchKit, HealthKit and SensorKit, geospatial analysis (ArcGIS, GeoPandas)

LANGUAGES AND CERTIFICATION

German (native), English (fluent), French (basic). Professional Scrum Master I (PSM I).